

Aditya Kulkarni

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kulkarniaditya.com | Raleigh, NC - Open to Remote / SF / NYC

PROFESSIONAL SUMMARY

Full-stack engineer with production experience across AI, distributed systems, and Web3. Built CryptoKnight - a trustless AI trading agent using ERC-4337 Account Abstraction and Session Keys, enabling 100% autonomous Polymarket farming without exposing private keys. At Isomer AI, shipped 10+ features across 7 deployments on TypeScript/Node.js/React with multi-model LLM integration (Claude, GPT-4o, Gemini). Built AlgoPulse handling 2,000 async AI inference requests/second with Temporal.io and Kafka. MS CS at NC State graduating May 2026 - drawn to Crypto x AI at the intersection of autonomous agents and on-chain security.

CORE COMPETENCIES

ERC-4337 Account Abstraction

Session Keys & Smart Accounts

AI Agent Architecture

TypeScript / Node.js / React

Python AI Agents

Multi-Model LLM Integration

Distributed Systems (Temporal + Kafka)

DeFi & Web3 (viem, wagmi)

Full-Stack Product Engineering

EXPERIENCE

Isomer AI

May 2025 - Present

Product Engineer Intern - San Francisco, CA (Remote)

- Shipped **10+ production features** across 7 deployments on TypeScript/Node.js/React/PostgreSQL - RBAC, real-time Pusher pipelines, document processing (PDF/email/S3), and multi-model AI agent tooling
- Built **MCP tool calling** across Claude, GPT-4o, and Gemini with Auth0, Redis, Docker, and Sentry - **394 commits** to the production codebase

NC State University

Apr 2025 - Present

Graduate Research Assistant

- Built a cloud-based genomic data platform (React + JBrowse 2) serving 100+ researchers across 500GB datasets on Google Cloud Run

Mphasis (HP Company)

Jun 2024 - Aug 2024

Software Developer Intern

- Built 5+ AI-driven workflows using OCR, NLP, and ML; automated deployments on AWS & GCP with Grafana monitoring

Capgemini

Jun 2023 - Aug 2023

AI/ML Intern

- Built Kafka streaming + Spark batch pipelines for real-time support ticket analysis; reduced latency by 20%

PROJECTS

CryptoKnight - Trustless AI Trading Agent

HackNC State · 2025

Solved the "Delegation Dilemma" in autonomous finance: an AI agent that farms Polymarket prediction markets **100% autonomously without ever holding private keys**. Used ERC-4337 Account Abstraction to issue Session Keys with strict on-chain policies - trade limits, max allowance, expiration, instant revocation. Funds stay in self-custody smart account vault at all times.

Alchemy Account Kit (ERC-4337) · Session Keys · Python agent · Sepolia · Next.js 16 · viem · wagmi

AlgoPulse - Agentic AI Platform

Independent · 2025

Event-driven AI orchestration platform handling **2,000 async inference req/sec**. Pivoted from polling to Supabase Edge Function streaming; Temporal.io for durable workflow state; Kafka for worker coordination and fault tolerance.

TypeScript · React · Node.js · Temporal.io · Kafka · Supabase Edge Functions

Glance AI - Privacy-First RAG System Independent · 2025

Compliance-ready RAG for fintech/healthcare with **per-tenant data isolation** at the vector retrieval layer - 0 cross-tenant context leakage by design.

Python · LLMs · Vector Databases · RAG

PerCent - GenAI Finance App HackNC24 Top 10 of 100+

Full-stack GenAI finance app with RAG-based chatbot (Qwen-14B-Chat) reducing manual financial research by 30%. Built complete MVP in 48 hours.

React · Hugging Face · Qwen-14B-Chat · Selenium

Mapify - ML & NLP Tool IEEE Published

T5-transformer pipeline converting 1,500+ PDFs/images into mind maps - 60% reduction in academic review time. Presented to 500+ at IEEE conference.

Python · T5 Transformer · NLTK · NetworkX

EDUCATION

North Carolina State University - MS Computer Science Aug 2024 - May 2026

Courses: Artificial Intelligence, Operating Systems, Data Science, Software Engineering, Algorithms

Mumbai University - BTech Computer Science Aug 2020 - May 2024

CGPA: 9.32/10 · Courses: DS & Algorithms, Machine Learning, Software Engineering, Big Data Infrastructure

SKILLS

Web3 / Blockchain: ERC-4337, Account Abstraction, Session Keys, viem, wagmi, Alchemy Account Kit, Solidity (basic), Sepolia

Languages: TypeScript, Python, JavaScript, Java, SQL, C, R

Frontend / Backend: React, Next.js, Node.js, Express, PostgreSQL, Redis, Prisma, REST APIs

AI/ML: OpenAI API, Anthropic Claude, Google Gemini, LangChain, MCP, RAG, Hugging Face, TensorFlow

Infra & Distributed: Temporal.io, Kafka, AWS, GCP, Docker, Kubernetes, CI/CD, Grafana